

PRST (31/08/2021).

Probability

Classical definition

$$\{H, T\} \quad P(H) = \frac{1}{2} \\ P(T) = \frac{1}{2}$$

$$\{1, 2, 3, 4, 5, 6\} \quad P(i) = \frac{1}{6} \quad i = 1, 2, 3, 4, 5, 6.$$

all outcomes are equally likely.

Empirical definition ———. & unfair coin.

$$P(H)$$

~~100~~ 100

N — number of times we toss the coin

$N(H)$ — number of times we get H out of N tosses.

$$\frac{N(H)}{N}$$

— relative frequency.

$$N = 100$$

$$\frac{N(H)}{N} = 0.65$$

$$N = 1000$$

$$\frac{N(H)}{N} = 0.689$$

$$N = 10^4$$

$$\frac{N(H)}{N} = 0.6998$$

$$N = 10^5$$

$$\frac{N(H)}{N} = 0.69999$$

$$N = 10^6$$

$$\frac{N(H)}{N} = 0.700100$$

subjective probability —

Axiomatic definition —

Sample space — collection of all possible outcomes of an experiment (Ω).

event — a subset of sample space Ω .

(Ex 2) $\Omega = \{1, 2, 3, 4, 5, 6\}$

$A = \{1\}$ event

$B = \{1, 6\}$ event

P : — we define probability as a real value function of events of Ω .

(i) $P(A) \geq 0$

(ii) $P(\Omega) = 1$

(iii) If $A_1, A_2, A_3, \dots$ are disjoint events then, ∞

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

$(A_i \cap A_j = \emptyset \text{ disjoint (mutually exclusive)})$

Result 1

proof.

$$P(\phi) = 0$$

$$A_1 = \phi, A_2 = \phi, \dots$$

$$\bigcup_{i=1}^{\infty} A_i = \phi$$

$$P(\phi) = P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i) = P(\phi) + P(\phi) + \dots$$

$$\Rightarrow P(\phi) = 0.$$

Result 2

$A_1, A_2, \dots, A_n$ are disjoint events

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i)$$

proof: $A_{n+1} = \phi, A_{n+2} = \phi, \dots$
 $A_i = \phi \quad i = n+1, n+2, \dots$

$$\bigcup_{i=1}^{\infty} A_i = \bigcup_{i=1}^n A_i \cup \phi \cup \phi \dots$$

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = P\left(\bigcup_{i=1}^n A_i\right) = P(A_1) + P(A_2) + \dots + P(A_n) + P(A_{n+1}) + P(A_{n+2}) + \dots$$

$$= \sum_{i=1}^{\infty} P(A_i)$$

Result 3

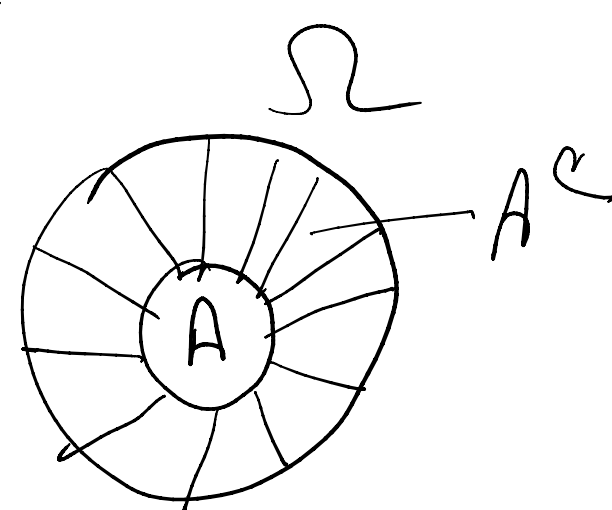
proof:

$$P(A^c) = 1 - P(A)$$

$$A \cup A^c = \Omega$$

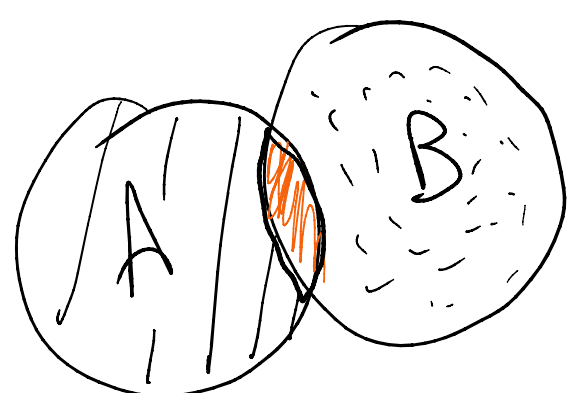
$$P(\Omega) = 1 = P(A \cup A^c) = P(A) + P(A^c)$$

$$\Rightarrow P(A^c) = 1 - P(A)$$



Result 6

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



$$(A - A \cap B) \cup (B - A \cap B) \cup (A \cap B)$$

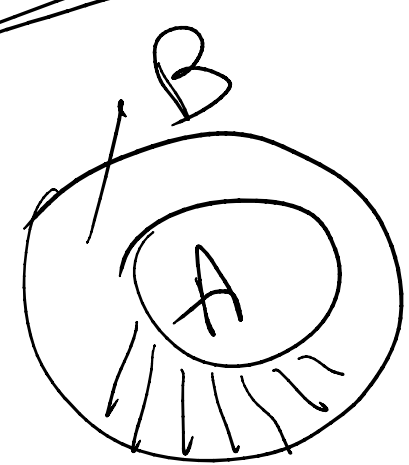
$$A \cup B = (A - A \cap B) \cup (B - A \cap B) \cup (A \cap B)$$

$$P(A \cup B) = P(A - A \cap B) + P(B - A \cap B) + P(A \cap B)$$

$$= P(A) - P(A \cap B) + P(B) - P(A \cap B) + P(A \cap B)$$

$$= P(A) + P(B) - P(A \cap B)$$

Result 5



$$A \subseteq B \Rightarrow P(A) \leq P(B)$$

$$B = A \cup (B - A)$$

$$P(B) = P(A) + \underbrace{P(B - A)}_{\geq 0}$$

$$P(B) \geq P(A)$$

Result

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i) - \sum_{i_1 < i_2} P(A_{i_1} \cap A_{i_2}) + \sum_{i_1 < i_2 < i_3} P(A_{i_1} \cap A_{i_2} \cap A_{i_3}) - \dots + (-1)^{n+1} P(A_1 \cap A_2 \cap \dots \cap A_n)$$

Result

$$P\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n P(A_i)$$

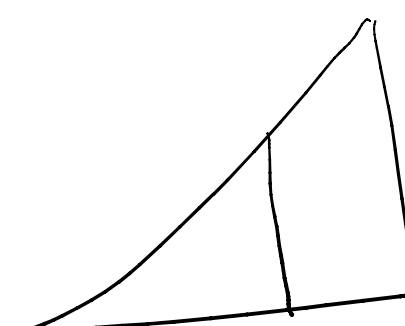
Tutorial-1

(Ex 5)

	freq	freq	Cumulative freq
0 — 60	$1000 \times \frac{4}{1000}$	40	40
60 — 62.5	$10^3 \times \frac{15-4}{100}$	110	150
62.5 — 95		700	850
95 — 100	$10^3 \times \frac{15-5}{100}$	100	950
≥ 100	$10^3 \times \frac{5}{100}$	50	1000

$$\frac{n}{2} = 500$$

$$\frac{850 - 150}{95 - 62.5} = \frac{500 - 150}{Med - 62.5}$$



$$Med = 78.75$$

$$1 \leq n \leq 2 - (\overline{r_k})^2$$

$$\frac{1}{n} \sum_{i=1}^n (\underbrace{x_i - \bar{x}})^2 = \frac{1}{n} \sum_{i=1}^n x_i^2 - (\bar{x})$$

$$\begin{aligned} & \frac{1}{n} \sum_{i=1}^n (x_i^2 + \bar{x}^2 - 2x_i\bar{x}) \\ &= \frac{1}{n} \sum_{i=1}^n x_i^2 + \bar{x}^2 - 2\bar{x} \left(\frac{1}{n} \sum_{i=1}^n x_i \right) \\ & \quad \quad \quad (2\bar{x}^2) \\ &= \frac{1}{n} \sum_{i=1}^n x_i^2 - \bar{x}^2 \end{aligned}$$

$$n \quad \text{Var} = 2S = \frac{1}{n} \sum x_i^2 - \bar{x}^2$$

$$\begin{aligned} \sum_{i=1}^n x_i &= 80 \\ n &= 32 \quad \bar{x} = \frac{80}{32} \end{aligned}$$

$$\begin{aligned} & \frac{1}{n} \sum_{i=1}^n (x_i - 4)^r \quad r = 1, 2, 3, 4 \\ & \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^r = \frac{1}{n} \sum_{i=1}^n (x_i - 4 + 4 - \bar{x})^r \end{aligned}$$

$$\frac{1}{n} \sum_{i=1}^n (x_i - 4) = -1.5$$

$$\Rightarrow \left(\frac{1}{n} \sum x_i \right) - 4 = -1.5$$

$$\bar{x} =$$

$$(d = \bar{x} - A)$$

$$m_1' = m_1 + d$$

$$m_2' = m_2 + 2m_1d + d^2$$

$$m_3' = m_3 + 3m_2d + 3m_1d^2 + d^3$$

$$m_4' = m_4 + 4m_3d + 6m_2d^2 + 4m_1d^3 + d^4$$